HOMEWORK 2
MATH 161

BLAKE FARMAN

Name: Solutions

1. If a rock is thrown upward on the planet Mars with a velocity of $10m/s$, its height in meters t seconds later is given by $y = 10t - 1.86t^2$.

(a) Find the average velocity over the given time intervals:

(i) $[1, 2]$

$$\frac{(10(2) - 1.86(2)^2) - (10(1) - 1.86(1)^2)}{2 - 1} = \frac{12.56 - 8.14}{1} = 4.42$$

(ii) $[1, 1.5]$

$$\frac{(10(1.5) - 1.86(1.5)^2) - 8.14}{1.5 - 1} = \frac{10.815 - 8.14}{.5} = 5.35$$

(iii) $[1, 1.1]$

$$\frac{(10(1.1) - 1.86(1.1)^2) - 8.14}{1.1 - 1} = 6.093\overline{9}$$

(iv) $[1, 1.01]$

$$\frac{(10(1.01) - 1.86(1.01)^2) - 8.14}{1.01 - 1} = 6.2613\overline{9}$$

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(v) $[1, 1.001]$

$$\frac{(10(1.001) - 1.86(1.001)^2) - 8.14}{1.001 - 1} = 6.27814$$

(b) Estimate the instantaneous velocity when $t = 1$.

We'll see soon that the exact answer is 6.28, but for now it seems reasonable to conclude that as the right endpoint tends towards 1, the average velocities tend toward a limiting value around 6.3.

2. Explain in your own words what is meant by

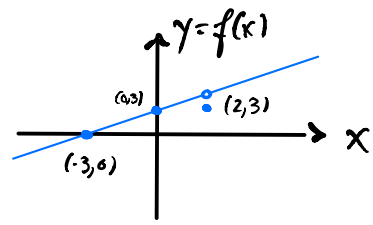
$$\lim_{x \to 2} f(x) = 5.$$

Is it possible for this statement to be true and yet $f(2) = 3$? Explain. [Hint: Drawing a picture might help.]

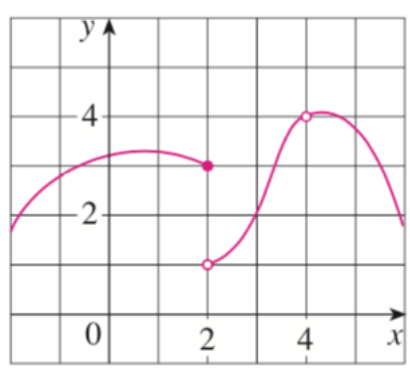
This means that whenever x is close to 2, $f(x)$ is close to 5.

Yes, it is possible for both statements to be true at the same time. An example is the piecewise defined function

$$f(x) = \begin{cases} 3 & x = 2, \\ x+3 & x \neq 2 \end{cases}$$



3. Use the given graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.



(a) $\lim_{x\to 2^-} f(x)$

3

(b) $\lim_{x\to 2^+} f(x)$

1

(c) $\lim_{x\to 2} f(x)$

This limit does not exist because the left- and right-hand limits are not the same.

(d) $f(2)$

3

(e) $\lim_{x\to 4} f(x)$

4

(f) $f(4)$

This value is not defined.